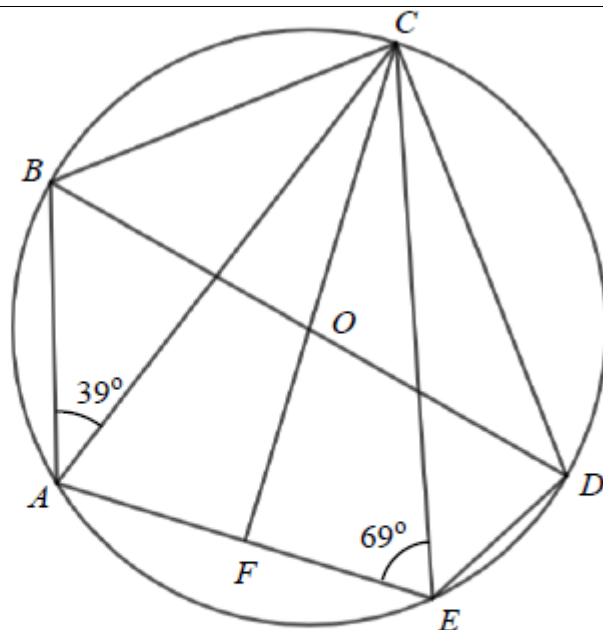




A, B, C, D and E are points on the circle, centre O .

CF bisects AE .

BOD is a diameter of the circle.

Angle $BAC = 39^\circ$ and $CEA = 69^\circ$.

(a) Find

(i) angle CDB ,

[1]

(ii) angle BCA ,

[2]

(iii) Angle ABD ,

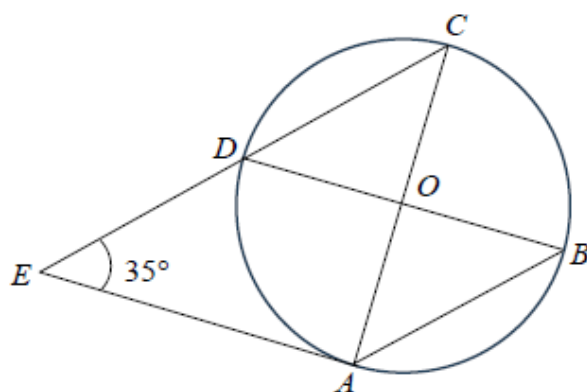
[3]

(iv) angle DCE .

[2]

(b) Given that the radius of the circle is 8 cm, find the area of the minor segment CGD .

[4]

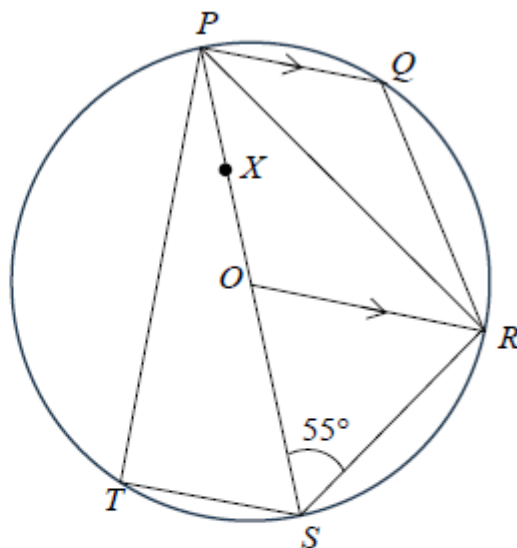


$ABCD$ is a circle with centre O .
 AC and BD intersect at O .
 The tangent at A meets CD produced at E .
 Angle $AED = 35^\circ$.

(a) Name a pair of congruent triangles and prove that they are congruent. [3]

(b) Find angle AOD .

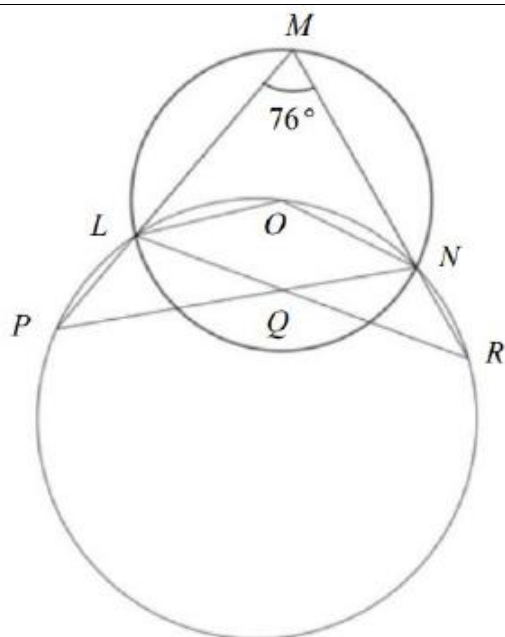
Show your working clearly and give a reason for each statement. [3]



$PQRST$ is a circle with centre O .
 PQ is parallel to OR .
 X is a point on PS such that $4PX = PS$.
 Angle $PSR = 55^\circ$.

(a) Find, giving a reason for each statement you make, angle PRQ . [4]

(b) Given that the area of triangle $SXT = 72 \text{ cm}^2$, calculate the area of triangle PXT . [2]



The diagram shows two circles with the bigger circle passing through the centre, O , of the smaller circle. The circles intersect at L and N . PLM and MNR are straight lines.

PN and LR intersect at Q . Angle $LMN = 76^\circ$.

Find, giving reason for each answer,

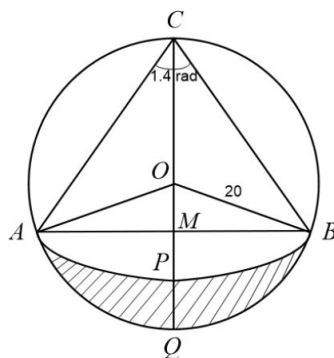
(a) angle QPL ,

[2]

(b) angle NQR .

[2]

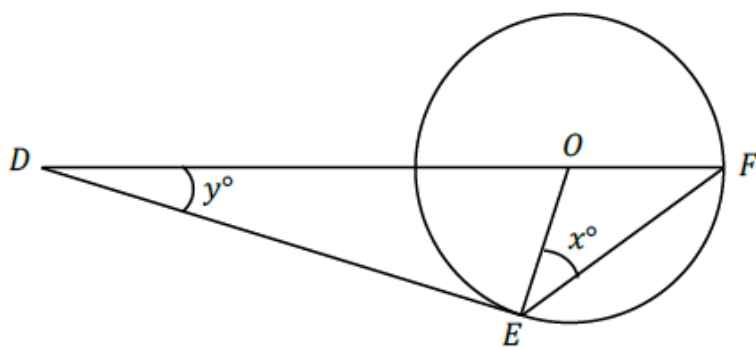
In the diagram, O is the centre of a circular picture $ACBQ$ of radius 20 m.
 APB is an arc of a circle centre C and radii AC and BC .
 COQ is a diameter of the circular picture $ACBQ$.
 COQ bisects AB at M .



If angle $ACB = 1.4$ radians,

- (a) show that AC is approximately 30.594 m, [3]
 (b) calculate the perimeter of the shaded region, [3]
 (c) calculate the total cost of painting the sector $OAQB$ if the painting cost is \$0.50 per square [3]

(a)



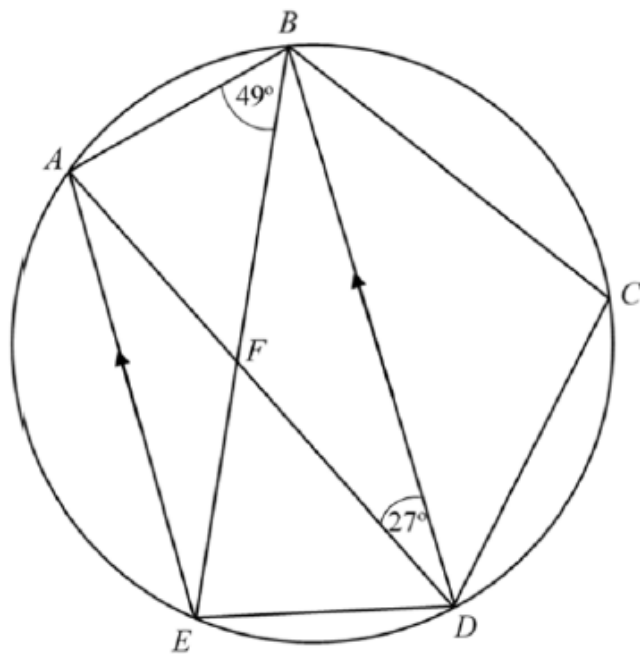
E and F are points on the circle, centre O .
 DOF is a straight line and DE is a tangent at E .
 Angle $FEO = x^\circ$ and angle $EDO = y^\circ$.

- (i) Show that $y = 90 - 2x$.
 State all the reasons clearly.
- (ii) Given that $y + x = 60$, find angle EOD .

[2]

[3]

- (b) The diagram shows a circle that passes through A, B, C, D and E .
 AFD and BFE are straight lines.
 AE and BD are parallel.
 Angle $ABE = 49^\circ$ and angle $ADB = 27^\circ$.



Find, give a reason for each step of your working,

- (i) angle AFE ,
- (ii) angle BCD .

[3]

[2]

1	(ai) $\angle CDB = 39^\circ$ (angles in same segment) [B1] (aii) $\angle ABC = 180^\circ - 69^\circ$ [M1] $= 111^\circ$ (angles in opp segment) $\angle BCA = 180^\circ - 111^\circ - 69^\circ$ (sum of angles in triangle) [A1] $= 30^\circ$ (aiii) $\angle CBD = 180^\circ - 90^\circ - 39^\circ$ $= 51^\circ$ (rt angle in semicircle) [M1 for 90°][M1] $\angle BCA = 111^\circ - 51^\circ$ $= 60^\circ$ [A1] (aiv) $\angle FCE = 90^\circ - 69^\circ = 21^\circ$ (perpendicular bisector of chord) [M1] $\angle DCE = 39^\circ - 21^\circ = 18^\circ$ (base angle of isosceles triangle) [A1] (b) $\angle COD = 180^\circ - 39^\circ - 39^\circ = 102^\circ$ (base angle of isos triangle) [M1] $\text{Area of segment} = \frac{102}{360} \times \pi \times 8^2 - \frac{1}{2} \times 8^2 \times \sin 102^\circ$ [M1][M1] $= 25.7 \text{ cm}^2$ [A1]
2	(a) $\left. \begin{array}{l} OD = OA \\ OC = OB \end{array} \right\}$ (radii of the circle) $\hat{C}OD = \hat{B}OA$ (vert. opp. angles) By SAS test, triangle COD and BOA are congruent. Alt $OD = OA$ (radii of circle) $\hat{D}CA = \hat{DBA}$ or $\hat{C}DB = \hat{CAB}$ (angles in same segment) $\hat{C}OD = \hat{B}OA$ (vert. opp. angles) By AAS test, triangle COD and BOA are congruent. (b) $\hat{E}AC = 90^\circ$ (radius perpendicular to tangent) $\hat{E}CA = 180^\circ - 90^\circ - 35^\circ$ (sum of angles in triangle) $= 55^\circ$ $\hat{A}OD = 2 \times 55^\circ$ (angle at centre = 2 x angle at circumference) $= 110^\circ$

3	(a) $\hat{PQR} = 180^\circ - 55^\circ$ (angles in opp. segments $= 125^\circ$ or opp. angles in cyclic quad.) $\hat{SRP} = 90^\circ$ (right angle in semicircle) $\hat{ORS} = 55^\circ$ (radii of circle, base angle of isosceles triangle) $\hat{ORP} = 90^\circ - 55^\circ = 35^\circ$ (right angle in semicircle) $\hat{QPR} = 35^\circ$ (alternate angles) $\hat{PRQ} = 180^\circ - 125^\circ - 35^\circ = 20^\circ$ (sum of angles in triangle) (b) $\frac{\text{Area of } \triangle PXT}{72} = \frac{1}{3}$ $\text{Area of } \triangle PXT = \frac{1}{3} \times 72 = 24 \text{ cm}^2$
4	(a) $\angle LON = 76^\circ \times 2 = 152^\circ$ (angle at center = 2 x angles at circumference) $\angle QPL = 180^\circ - 152^\circ$ (angles in opposite segment) $= 28^\circ$ (b) $\angle PNR = 76^\circ + 28^\circ$ (exterior angle of triangle) $= 104^\circ$ $\angle NRL = 28^\circ$ (angles in same segment) $\angle NQR = 180^\circ - 104^\circ - 28^\circ$ ($\angle$ sum of Δ) $= 48^\circ$

5

(a)

$$\angle AOB = 2(1.4) \quad (\angle \text{ at centre} = 2\angle \text{ s at circumference})$$

$$= 2.8 \text{ radians}$$

$$\angle AOC = (\pi - 1.4) \text{ radians} \quad (\text{Adj } \angle \text{ s on a st. line})$$

$$AC^2 = 20^2 + 20^2 - 2 \times 20 \times 20 \times \cos(\pi - 1.4)$$

$$AC \approx \sqrt{935.9737}$$

$$\approx 30.5936$$

$$\approx 30.594 \text{ m (Shown)}$$

(b)

$$\text{Perimeter} = \text{Arc length } APB + \text{Arc length } AQB$$

$$= 30.594 \times 1.4 + 20 \times 2.8$$

$$= 98.831$$

$$\approx 98.8 \text{ m}$$

(c)

$$\text{Area of sector } OAQB = \frac{1}{2} \times 20^2 \times 2.8$$

$$= 560 \text{ m}^2$$

$$\text{Total cost} = \$ (0.50 \times 560)$$

$$= \$280$$

6

(ai)

$\angle OFE = x^\circ$ (Base angles of isosceles triangle since OE and OF are radii of same circle.)

$\angle OED = 90^\circ$ (Tangent to a circle)

$y + 90 + x + x = 180$ (Angles sum of triangle)

$\therefore y = 90 - 2x$ (Shown)

(aii)

$y = 90 - 2x \dots (1)$

$y + x = 60. \dots (2)$

Substitute (1) into (2) :

$$90 - 2x + x = 60$$

$$x = 30, y = 30$$

$$\therefore \angle EOD = 90^\circ - y^\circ$$

$$\therefore \angle EOD = 90^\circ - 30^\circ = 60^\circ$$

(bi)

$$\angle AEB = \angle ADB = 27^\circ$$

(Angles in the same segment)

$$\angle EAD = \angle ADB = 27^\circ$$

(Alternate angles, BD parallel AE)

$$\angle AFE + \angle AEB + \angle EAD = 180^\circ$$

(Angles sum of triangle)

(bii)

$$\angle AFE = \angle DAB + \angle ABE \text{ (Exterior angle)}$$

$$\angle DAB = 126^\circ - 49^\circ = 77^\circ$$

(Opposite angles of a cyclic quadrilateral)

$$\angle BCD = 180^\circ - \angle DAB$$

$$= 180^\circ - 77^\circ$$

$$= 103^\circ$$